CPR/BLS in Office Practice

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Definitions

- ☑ CPR—Cardiopulmonary resuscitation
- ☑ BLS—Basic life support—sequence of steps to perform CPR. This term is promoted by the American Heart Association (AHA).

Who Would Need Cardiopulmonary Resuscitation?

Any victim of sudden cardiac arrest would require CPR.
Cardiac arrest may occur in a variety of situations, both in- and out-of-hospital circumstances.

Causes

Physiological derangements of:

- ☑ Circulation (shock) and/or
- ☑ Respiration (respiratory insufficiency) and/or
- ☑ Brain (cortical or brainstem dysfunction). Either suddenly or due to underlying infections, injuries, toxins, animal bites, or comorbidities result in cardiac arrest situations in any age group.

Sudden cardiac arrest in adults is often due to cardiac causes including rhythm abnormalities like ventricular fibrillation or pulseless ventricular tachycardia. On the other hand, major cause of cardiac arrest in children is due to respiratory failure and/or shock.

A victim of cardiac arrest has the best chance of recovery, if the CPR is administered by the person standing next to the victim immediately on recognition of the condition.

How to Identify Cardiac Arrest?

A victim is in cardiac arrest if he/she suddenly becomes:

- ☑ Unconscious and unresponsive,
- ☑ Stops breathing or is gasping with ineffective breaths and
- ☑ Does not have a pulse.

A child is labeled as "in cardiac arrest" even if there is a pulse beating below or up to 60 bpm *without features of circulation* along with "unresponsiveness" and "ineffective or not breathing".

Assessment of a Victim of Cardiac Arrest

Three things you should look for to diagnose cardiac arrest.

Unresponsiveness

"Firmly" tap the victim on shoulder **(Fig. 1)** and ask a question "loudly" in a language he/she would understand like—Are you okay? or How are you? or Do you need some help?

- ☑ It usually takes <5 seconds to elicit (un)responsiveness of a victim. One expects the small child to cry or react in response to the tap, if not unresponsive.
- ☑ If there is no response, check for the surroundings. Remove the victim to a safe area, if there is an imminent danger to any life such as fire, smoke, electric current, and building collapse.

Fig. 1: Assess responsiveness by tapping shoulder.

Breathing Assessment

Check breathing simultaneously if the victim is unresponsive.

- ☑ Look and listen while bringing your ears close to the mouth of the victim observing the chest wall movements at the same time.
- ☑ "Open the airway" in case the neck is too flexed or extended to appreciate breathing.

Fig. 2: Opening the airway.

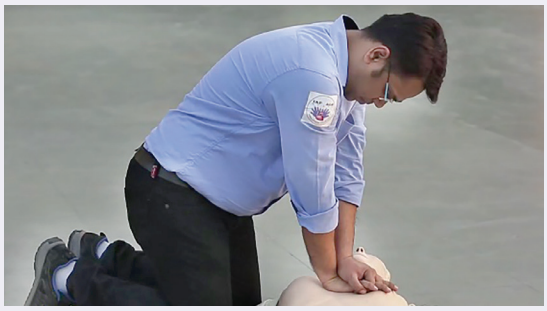
Opening of the airway requires tilting the victim's neck by placing the ulnar side of your one hand on the forehead of the victim and lifting his/her jaw by holding on to the mandible with the thumb and index finger of your other hand ***(Fig. 2)***.

- ☑ One should not take >5–10 seconds to make an assessment of breathing.
- ☑ There may be three situations—the victim might be breathing normally, or not breathing at all or might be gasping, i.e., ineffective breathing efforts—not sufficient for adequate inflow of air in lungs. The latter two conditions—not breathing and gasping are dealt as "not breathing".

Assessment of Pulse

- ☑ Presence (or its absence) is checked in the carotid area in adults.
- ☑ In children the pulse is checked in the femoral areas and in infants it is checked in brachial areas.
- ☑ Palpating a pulse with confidence in an emergency is not easy even for a trained person. CPR should not be delayed by >10 seconds while checking for pulse in an unresponsive and not breathing victim.

Steps of Cardiopulmonary Resuscitation

Upon identifying unresponsiveness and not breathing, call for help. Ask for automated external defibrillator (AED), if available **(Fig. 3)**. **Fig. 3:** Call for help and ask for AED, if available.	Use local numbers of nearby emergency care facilities or 112/108 as applicable in your area. Take help from bystanders to call services.
Place the victim on a flat firm surface; kneel down on your knees beside the victim.	Use a CPR board under the mattress, if the victim is on a bed.
Undress the victim and place the heel of your dominant hand over the lower half of the sternum of the victim **(Fig. 4)**. Place your other hand on your first hand, grasping the fingers of first hand by second hand. **Fig. 4:** Location of chest compression.	Ensure a female attendant along with you when helping a female victim.
Start chest compressions. Count loudly while providing chest compressions **(Fig. 5)**.  **Fig. 5:** Start chest compression with proper technique.	☑ Keep your elbows straight ☑ Your shoulders should be vertically above the chest of victim ☑ Push hard up to 5–6 cm ☑ Push fast @100–120 bpm

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Avoid interruptions in chest compression.	Allow full chest recoil.
Provide two breaths after every 30 chest compressions. Provide breathing by "mouth-to-mouth" or "mouth-to-mask" or bag-valve-mask device **(Fig. 6)**. **Fig. 6:** Use of one-way valve mask for breathing. In case pulse is palpable and victim is not breathing, provide breaths @10–12 per minute in adults without chest compressions (rescue breaths).	Breaths should be slow and sufficient to just raise the chest as during normal breathing.
☑ Apply AED pads as soon as available. ☑ Switch on the AED. ☑ Stop chest compressions to analyze rhythm. Press the shock button on AED, if AED prompts so. ☑ Resume CPR immediately after shock is delivered.	☑ AED may not prompt for shock in all victims, depending upon the rhythm. ☑ If shock is advised, take all precautions to prevent injury to yourself and your team members by staying away from the victim while shock is delivered.
☑ Switch roles of compressor and ventilator every 2 minutes. Check pulse every 2 minutes. ☑ CPR is provided till the pulse returns or victim is handed over to advanced life support (ALS) team and/or shifted to intensive care unit (ICU).	Plan to shift to ICU early
Steps are different for child CPR	
☑ Chest compression may be done by one hand only. ☑ In a small child, the chest should be compressed up to one-third of anteroposterior (AP) diameter. ☑ Compression ventilation ratio should be 15:2 for children and infants. If you are not trained in pediatric CPR, give chest compressions and ventilations @ 30:2, like adults. ☑ Rescue breathing administered to child with pulse above 60 bpm **(Fig. 7)**, but unresponsive and not breathing would be of lesser volume @ 15–20 breaths per minute. ☑ If the AED paddles for children are not available, then those for adults are used placing them anteriorly and posteriorly on the child's/infant's chest.	Child is between 1 year of age and puberty. Any individual with features of breast development or axillary hairs is managed as adult as above.

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Steps of Cardiopulmonary Resuscitation

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Steps different for an infant	***Child up to 1 year of age***
Chest compressions may be done by either two fingers over the sternum—just below the intermammary line or by two thumbs on sternum with hands encircling the chest wall of the infant **(Fig. 8)**.	Single rescuers should use two-finger technique though two-thumb technique is a shade better.

Fig. 7: Checking pulse in brachial area in an infant.

Fig. 8: Two-thumb encircling technique of chest compression in infant.

Complications of Cardiopulmonary Resuscitation

- ☑ Fracture of ribs or the sternum may occur, if chest compressions are not performed correctly; however the incidence of increased mortality from such fractures is low as compared to benefits of CPR. One must learn the proper technique.
- ☑ *Aspiration and vomiting*: The most common problem encountered during CPR. If the secretions are not cleared properly and immediately, the victim is likely to aspirate it into his lungs leading to aspiration pneumonia.
- ☑ *Abdominal distension*: Due to inappropriate ventilation by bag and mask. Again, using proper technique is the key to avoiding complications.

Cardiopulmonary resuscitation has been described as "violent, painful, and undignified" in the past. With increasing public awareness and its widely documented benefits, one has to provide CPR in indicated patients as it provides a four-fold increase in survival rate, even if it is performed by bystanders.

Conclusion

Cardiopulmonary resuscitation/basic life support is an essential skill that every healthcare provider must master to support life immediately, if a patient or victim collapses or develops a cardiac arrest. The Indian Academy of Pediatrics through its IAP ALS BLS group trains healthcare professionals and lay persons through its variety of modules suiting individual requirements **(Fig. 9)**.

We also have an "e-Sanjeevani" module for learning the steps of CPR at our convenience accessible at *https://forms.gle/jpZnatDQMNUGgGt76* free of cost. This may be followed by practicing skills on mannequins supervised by trained instructors at IAP accredited CPR centers spread across the country after which a completion certificate is also available.

Contact zone coordinators of IAP ALS BLS group of your geographical IAP zone through the central IAP office for organizing the course for your colleagues or friends.

Fig. 9: Indian Academy of Pediatrics basic life support (IAP BLS) training session providing one-to-one training.

Further Reading

- ☑ Garg R, Ahmed SM, Kapoor MC, Mishra BB, Rao SC, Kalandoor MV, et al. Basic cardiopulmonary life support for resuscitation by trained paramedics and medics outside the hospital. Indian J Anaesth. 2017;61:874-82.
- ☑ Taneja LN, Tiwari L. BLS for Professionals, 3rd edition. Indian Academy of Pediatrics ALS BLS Group; 2018.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

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